

Chika Inagaki

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EDUCATION

University of California, San Diego

Sep. 2024 – Present

B.S. Electrical Engineering, Minor in Cognitive Science

La Jolla, CA

- GPA 4.0 — Provost Honors every quarter

EXPERIENCE

Student Researcher – Multimodal Neural Sensing

Dec. 2025 – Present

Integrated Systems Neuroengineering Lab (ISNL), PI: Gert Cauwenberghs, UC San Diego

- Fully designed experimental protocols to measure fatigue with controlled cognitive tasks, such as Mackworth clock
- Implemented Python pipelines for multimodal data collection, including eye-tracking on Meta glasses Project Aria, EEG with Patch EEG, and standardized sleepiness surveys like KSS & ESS
- Developed a multimodal reinforcement learning model to detect fatigue on a continuous time scale
- Managed and helped other undergraduate research assistants with an interactive interview-based task

Student Researcher – Neuromorphic Hardware

Sep. 2025 – Present

Integrated Systems Neuroengineering Lab (ISNL), PI: Gert Cauwenberghs, UC San Diego

- Characterized selector-less RRAM memristor crossbar arrays by measuring read accuracy, resistance distributions, and array-level conductance behavior
- Built Python-based test and analysis scripts to optimize readout accuracy across multiple resistance states
- Conducted Hadamard-Walsh decoding method to read out array-parallel conductance

Undergraduate Teaching Assistant – ECE 5 (Introduction to ECE)

Sep. 2025 – Present

UC San Diego

- Co-developed and supported a hands-on embedded systems project (photoresistor-based line-following robot) including analog sensing, motor control, and PID tuning
- Provided one-on-one and small-group guidance on circuit analysis, debugging, and measurement techniques
- Reviewed student work and ensured lab procedures and safety practices were followed

ENGINEERING PROJECTS

Smart Reminder Pill Case for Elderly (Personal Project)

Feb. 2025 – Present

- Led hardware circuit design and soldering for a fully-hardware pill case with different types of pills in each case, with a modular style
- Designed and assembled a portable embedded system using Arduino Nano, sensors, keypad, LCD, buzzer, and LEDs, including system-level integration and debugging
- Advanced the project further toward deployment for the elderly not familiar with phone apps through materials research and market analysis through interviews at hospitals in the US and Japan

EMG-Controlled Prosthetic Hand (Triton Neurotech)

Oct. 2024 – Feb. 2025

- Analyzed EMG signals through patch electrodes with MATLAB signal processing toolbox to detect hand motion
- Designed and iterated mechanical components using CAD-designed fingers, strings, servos, and Raspberry Pi 5
- Developed a prosthetic hand that mimics the motion of the user's hand

TECHNICAL SKILLS

ML: Deep learning fundamentals, reinforcement learning, hyperdimensional computing **Languages:** Python, C/C++, MATLAB, Java

Hardware & Systems: Embedded systems, FPGA-based control, PID control, Verilog, LTSpice, CAD

Signal Processing: EEG/EMG analysis, multimodal sensing, basic feature extraction

AWARDS

EN Foundation Full-Ride Scholarship (2 selected from 250+ applicants), 2024–2028

DigiKey Award, Hard Hack 2025 Smart Reminder Pill Case

Overall 3rd Place, Diamond Hacks 2025 Volunteering website for high school students